

DDTA - Scheda di lettura compilata

SRC-0044 - BA0-R systems-modeling prior art

REVIEW-ONLY: da approvare prima del caricamento nel repository.

Identita bibliografica e accesso

Source ID	SRC-0044
Titolo	The Architecture Analysis & Design Language (AADL): An Introduction
Autori/ente	Peter H. Feiler; David P. Gluch; John J. Hudak
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1. Research problem and contribution

Introduces AADL as a language with precise semantics and extensible properties for early/repeated architecture analysis, especially real-time and high-dependability systems. It models software, computational hardware and composite system components and supports mappings between application and execution platform.

2. Starting artifacts and generated representations

AADL component types expose features/ports; implementations contain subcomponents and connections. Execution-platform abstractions include processor, memory, bus and device. Modes represent configuration variants. Flow declarations and end-to-end flows are distinct from connections. Properties attach analysis-relevant values to components, ports and connections.

3. Traceability, change, automation and human role

AADL's semantics are analyzability-oriented and permit partial specifications and repeated refinement. The report does not provide DDTA-style document provenance or governance; its contribution is a formal architecture representation that analysis tools can consume.

4. Findings, limitations and relationship with DDTA

It supports a generic "connection with properties" notion more strongly than a universal Channel root. Bus is a specialized communication platform component; memory is a specialized execution-platform component. Therefore Channel and Store are real system concerns but not necessarily top-level ontology primitives. It also shows that useful analysis can start before full implementation detail exists.

Evidenza utile per BA0/BA1

Pag.	Ruolo	Sezione	Interpretazione DDTA
7	foundation	3 AADL Language Abstractions	An analyzable core can be abstract while retaining formal semantics.
21	foundation	4.10-4.11 Partial Specifications	Base Analysis need not require a fully detailed implementation model before analysis can begin.
44	challenge	6.2 Memory	Storage need does not imply Store must be a universal root type.
46	challenge	6.3 Bus	Channel may be a role/specialization of a generic connection/resource rather than a base root.
56	challenge	8 Component Interactions	A single Interaction abstraction risks erasing analysis-relevant distinctions.

Pag.	Ruolo	Sezione	Interpretazione DDTA
58	foundation	8.1.2 Port Connections	Connection and endpoint identity are analyzable first-class concerns.
86	foundation	9 Modes	State/mode can materially change analyzable architecture.
91	foundation	10 Flows	End-to-end flow and structural connection are distinct semantic layers.
95	foundation	11 Properties	Properties/constraints may be preferable to multiplying structural metaclasses.

Bibliography mining / follow-up

- SAE AS5506 AADL standard
- OSATE/AADL analysis literature
- AADL Error Model Annex as later analysis-overlay comparison (not for BA core)

Governance note

Questa scheda è un ausilio di review. Le future citazioni di tesi devono risolversi alla fonte registrata e a una posizione esatta nell'excerpt ledger approvato. Nessun concetto esterno diventa automaticamente un BAE DDTA.